

Full length article

Preparation of magnetic resin microspheres M-P(MMA-DVB-GMA) and the adsorption property to heavy metal ions

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ARTICLE INFO

Keywords:

Magnetic resin microspheres
Heavy metal ions
Adsorption
Chromium ion
Lead ion
Copper ion

ABSTRACT

Magnetic resin microspheres called MPMDG₂ were prepared by suspension polymerization combined with basic and acidic hydrolysis. The adsorption property of resin microspheres to heavy metal ions was investigated. Structural and morphological analysis revealed that the resin had high specific surface area and mesoporous structure. The resin contained about 9.2 wt% Fe₃O₄ and exhibited a high magnetic response. Adsorption experiments showed that the resin can selectively absorb Cr(III) and Pb(II) ions. The adsorption process was well described by a pseudo-second-order kinetic model, and the equilibrium adsorption data fitted the Langmuir isotherm model better than the Freundlich model. Ion exchange along with chemistry chelating was the most probable mechanisms for the adsorption of metal ions on the resin. The superior magnetic response, regeneration and selective adsorption property of the MPMDG₂ make it a good candidate as an adsorbent for removing Cr(III), Pb(II) and Cu(II) from wastewater.

1. Introduction

Heavy metal ions (HMIs) in wastewater pose a serious threat to human health and natural environment because of the high toxicity, non-biodegradability and bioaccumulation characteristics [1]. HMIs like lead (Pb), cadmium (Cd), copper (Cu) and chrome (Cr), come from chemical manufacturing, battery manufacturing industries, metallurgical, leather tanning and mining [2]. At present, obvious chemical and physical methods have been developed to remove HMIs from wastewater. Comparatively, chemical methods e.g. chemical precipitation and electrochemical reduction are effective in removing high concentration of HMIs from wastewater, but could cause secondary pollution due to the use of chemical reagents [3]. Physical methods such as membrane separation, adsorption and ion exchange are feasible to remove low concentration of HMIs from wastewater, and environmental-friendly and easy to handle [4]. Obvious adsorbents including active carbon, carbon nanotube and resins have been developed [5,6]. Among them, resins show great advantages due to its high specific surface area, mechanical strength, adsorption capacity, easy regeneration and low cost [7]. Above all, ion-exchange resins give the best effect on the treatment of heavy metal waste water [8].

Conventional sulfonic acid ion exchange resin exhibits high activity to remove HMIs, but the adsorption selectivity on the resin is low, and furthermore, the removal of HMIs is easily disturbed by the coexisting

ions. Recent literatures reported that novel chelating resins functionalized with OH, NH₂, COOH and SH groups showed an improved adsorption selectivity for Hg(II), Pb(II) and Cu(II) [9,10]. Methyl methacrylate (MMA) was often selected as monomer for preparation of polymers because alkaline hydrolysis of the methyl ester in MMA easily produced COOH on the surface of polymers [11,12]. Recently, the application of magnetic polymer adsorbents for the removal of HMIs has become a new topic of research since it combines the advantages from both components: magnetic particles and polymers. The magnetic particles make the rapid and facile separation of adsorbents possible in an external magnetic field. The high specific surface area and functional groups of polymers help to adsorb HMIs on resins [13]. For instance, a novel magnetic cation-exchange resin developed via suspension polymerization (methyl acrylate as monomer, diallyl itaconate and divinylbenzene as cross-linkers) and basic hydrolysis process, exhibited high adsorption capacities for Cu(II) and Ni(II) [14]. According to Kara et al., magnetic vinylphenyl boronic acid microspheres were prepared by copolymerizing of ethylene glycol dimethylacrylate with 4-vinyl phenyl boronic acid, showing great potential for Cr(VI) removal [15]. In another report, micron-sized monodisperse superparamagnetic polyglycidyl methacrylate particles with functional amino groups were prepared by the dispersion polymerization and subsequently chemical modification, and further direct precipitation of magnetite nanoparticles within the polymer particles [16]. In this study, glycidyl

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Received 18 January 2019; Received in revised form 28 June 2019; Accepted 15 August 2019

Available online 16 August 2019

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methacrylate (GMA) was selected as monomer because its epoxy groups can be easily converted into alcohol functional group by acidic hydrolysis. Therefore, the development of magnetic polymers with high concentration of functional groups such as COOH and OH via selecting proper monomers and technical process for selective adsorption of HMIs from wastewater is possible.

In present work, magnetic resin microspheres with carboxyl and hydroxyl groups were prepared by suspension polymerization combined with acidic and basic hydrolysis. The adsorption property to HMIs was investigated.

2. Experimental

2.1. Materials

Methyl methacrylate (MMA, AR), divinylbenzene (DVB, 55%), glycidyl methacrylate (GMA, AR) and dibenzoyl peroxide (BPO, AR) were obtained from Shanghai Mclean Biochemical Technology Co. Ltd., China. $\text{FeCl}_3 \cdot 6\text{H}_2\text{O}$, $\text{FeCl}_2 \cdot 4\text{H}_2\text{O}$, polyvinyl alcohol (PVA) and polyvinylpyrrolidone (PVP) were of analytical grade and purchased from Guangzhou Jinhua Da Chemical Reagent Co., Ltd. Oleic acid (OA), toluene and heptanes were of analytical grade and obtained from Chengdu Kelong Chemical Reagent Factory.

2.2. Preparation of magnetic resin microspheres

M-P(MMA-DVB-GMA) magnetic resin microspheres were synthesized by suspension polymerization combined with basic and acidic hydrolysis. Typically, the organic phase contained 1.5 g OA- Fe_3O_4 [17], 5.0 g MMA, 1.0 g GMA, 1.0 g DVB (as cross-linker) and 0.1 g BPO (as initiator) mixed together with 4 mL toluene and 3 mL *n*-heptane. The aqueous phase contained 100 mL water, 0.8 g PVA, 0.8 g PVP and 4.0 g NaCl. The organic phase was supersonic vibrated for 30 min, and then mixed with the aqueous phase in a stainless steel reactor, and further programmed heated to 70 °C at a heating rate of 1 °C/min. The polymerization reaction was maintained at 70 °C for 2.5 h and then at 80 °C for 2.5 h under stirring. The resultant solid sample was alternately washed with hot and cool water, and then vacuum dried for 16 h. The precursor (2 g) was dispersed in 50 mL 10% NaOH aqueous solution and hydrolyzed at 40 °C for 16 h (basic hydrolysis), and the obtained sample was denoted by MPMDG₁. 2 g MPMDG₁ was dispersed in 50 mL 0.2 M H_2SO_4 aqueous solution and hydrolyzed at 80 °C for 8 h (acidic hydrolysis), and the obtained resin sample was denoted by MPMDG₂. The polymerization scheme for the preparation of M-P(MMA-DVB-GMA) is shown in Fig. 1.

2.3. Characterization

The obtained resins were characterized by Fourier transform infrared (FTIR) spectroscopy (8400S, Shimadzu, Japan), scanning electron microscopy (SEM) (S-3400 N, Hitachi, Japan), powder X-ray

diffraction (XRD) (SmartLab3, Rigaku, Japan), Physical Property Measurement System (PPMS-9, Quantum Design, USA), Brunauer-Emmett-Teller (BET) surface area measurements (NOVA 2200e, Quantachrome, USA). The adsorption behavior for HMIs on the obtained resins was characterized by inductively coupled plasma optical emission spectroscopy (ICP – OES, Agilent 7700e, USA).

2.4. Adsorption of HMIs

The actual application of resin microspheres was demonstrated through the adsorption of Cu(II), Cd(II), Pb(II) and Cr(III) ions. The stock solutions (1000 mg/L) of Cu(II) and Pb(II) salts was prepared by dissolving metal copper and lead with nitric acid, and then diluted in water. The stock solution of Cd(II) salt was obtained by dissolving CdO with 6 M HCl and diluted in water. The Cr(III) stock solution was provided by Shangdong Xiya Chemical Co., Ltd. The stock solutions were diluted to different concentrations using deionized water to obtain simulated wastewater. Batch experiments were conducted using 40 mL of the metal ions mixed solution (the concentration of Cu(II) and Cd(II) is 100 mg/L; the concentration of Cr(III) and Pb(II) is 200 mg/L) to determine the effect of pH and contact time on the adsorption of HMIs. After the pH of the solution was adjusted to a desired value using both 1 M HCl and 1 M NaOH, 60 mg of the resin (MPMDG₂) was added to the solution. The resultant suspensions were agitated for a specified time by a shaker, afterwards, the filtrate was analyzed by ICP-MS. Then, the effect of the initial concentration of metal ions on the adsorption performance of the resin was investigated at pH 5.5 for a contact time of 8 h using 40 mL of the metal ions mixed solution with different initial concentrations (20, 50, 80, 110, 140, 170, 200 and 230 mg/L). The adsorption capacity (*Q*) and removal efficiency (*E*) were calculated according to Eq. (1) and (2), respectively:

$$Q(\text{mg/g}) = \frac{V(C_0 - C_e)}{M} \quad (1)$$

$$E(\%) = \frac{(C_0 - C_e)}{C_0} \times 100\% \quad (2)$$

where *V* is the volume of the metal ion mixed solution (L), *M* is the weight of the resin (g), *C*₀ and *C*_e are the initial and final equilibrium concentrations of metal ions in the solution (mg/L), respectively.

2.5. Desorption and regeneration experiments

To investigate the reusability of the obtained resin, desorption and regeneration experiments were conducted at room temperature (RT). For the desorption studies, the metal ions-loaded resin was dispersed in 50 mL of 0.1 M HCl solution and left to stand for 8 h. After desorption, the resin was repeatedly washed with deionized water and then dried for reuse. Six consecutive adsorption-desorption cycles were conducted, and in each cycle, the initial concentration of metal ions in the mixed solution was: Cu(II) and Cd(II) at 100 mg/L, and Cr(III) and Pb(II) at

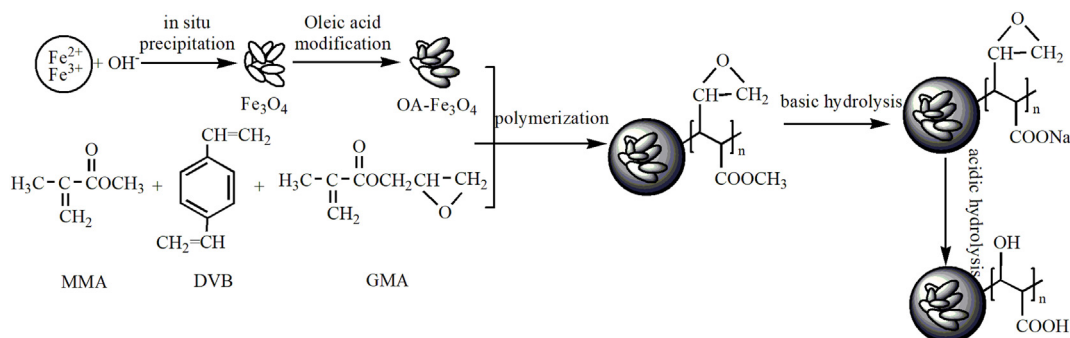


Fig. 1. Schematic diagram for preparation of M-P(MMA-DVB-GMA).

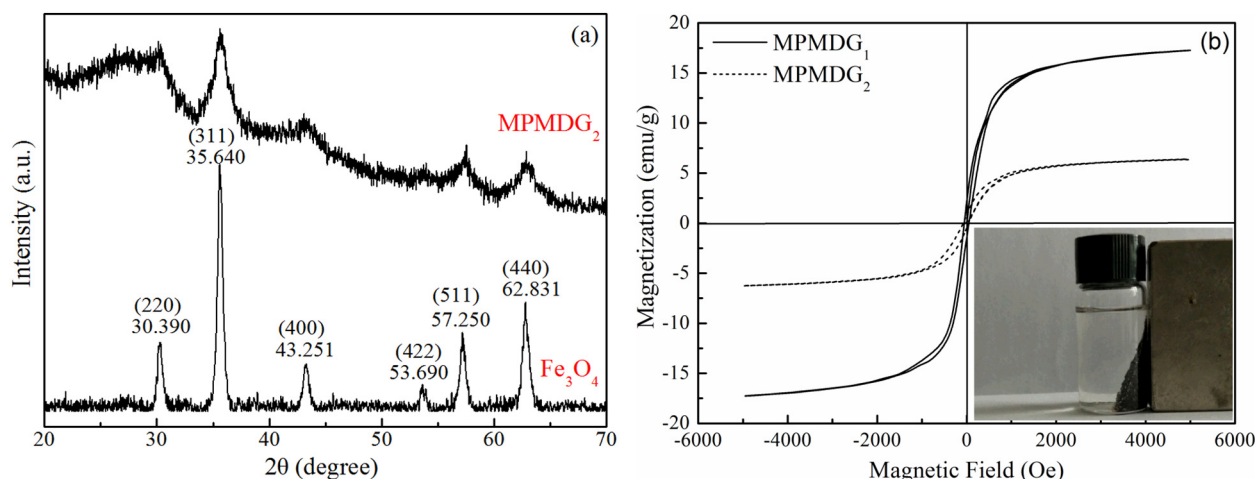


Fig. 2. (a) X-ray powder diffractogram of MPMDG₂, (b) magnetic hysteresis loops of magnetic resin particles.

200 mg/L, resin dosage was 60 mg and the mixed solution was 40 mL. The adsorption-desorption cycle efficiency (R) was calculated using Eq. (3).

$$R(\%) = \frac{Q_n}{Q_0} \times 100\% \quad (3)$$

where Q_0 is the initial adsorption capacity of the resin (mg/g), and Q_n is the adsorption capacity of the resin after desorption for n -times.

3. Results and discussion

3.1. Characterization of the resin microsphere

The XRD pattern of the MPMDG₂ is presented in Fig. 2(a), which is compared with that of pure Fe₃O₄ particles. Obvious diffraction peaks appear at $2\theta = 30.390^\circ$, 35.640° , 43.251° , 57.250° and 62.831° , which correspond to the planes (220), (311), (400), (511) and (440), respectively, of the inverse spinel structure of Fe₃O₄ (JCPDS file no. 16-629) [18]. The magnetization curves of resin samples measured at RT are shown in Fig. 2(b). The loop areas for both resin samples (MPMDG₁, MPMDG₂) are narrow, indicating that Fe₃O₄ particles in resins are superparamagnetic, and have a particle size smaller than 20 nm [19]. The saturation magnetizations (M_s) of MPMDG₁ and MPMDG₂ are 17.26 and 6.29 emu/g, respectively. Therefore, the contents of Fe₃O₄ in MPMDG₁ and MPMDG₂ were estimated to be 25.3 wt% and 9.2 wt%, respectively, which were calculated by comparison of the M_s of resin samples to that of pure superparamagnetic Fe₃O₄ particles (67.3 emu/g) [20]. The inner illustration in Fig. 2(b) indicates that the resin particles can be easily separated by exposing them to an external magnetic field.

Fig. 3 shows the SEM images of the obtained resin microspheres. Both MPMDG₁ and MPMDG₂ exist as spherical beads, with grain sizes of approximately 150–260 μm . By contrast, the MPMDG₂ present a rougher surface and more obvious porous structure. As shown in Table 1, the average pore diameters of the MPMDG₁ and MPMDG₂ are 4 and 4.6 nm, respectively, indicating that the resulting resins are mesopore-predominant materials. OA-Fe₃O₄ can combine with MMA, DVB and/or GMA via polymerization reaction with the double bonds, as indicated in Fig. 1, while the Fe₃O₄ particles and the long aliphatic chains of oleic acid could weaken the polymerization reaction between the three monomers (MMA, GMA and DVB), leading to the formation of mesoporous structure. After acidic hydrolysis, specific surface area decrease, accompanied with slight increase in the total pore volume and average pore diameter, which would be due to partially removal of Fe₃O₄ particle from resin matrix and opening the epoxy group in GMA. The high specific surface area and mesoporous properties would be

favorable for diffusion of metal ions and their adsorption on the resins.

In order to find out the effect of addition of Fe₃O₄ on polymerization reaction, Fe₃O₄-free resin microspheres were prepared according to the similar methods to that of MPMDG₁ and MPMDG₂ except without addition of OA-Fe₃O₄. Correspondingly, PMDG₁ and PMDG₂ were obtained from basic hydrolysis and two-step hydrolysis (basic and acidic hydrolysis) of P(MMA-DVB-GMA) microspheres, respectively. Fig. 4 shows the FTIR spectra of PMDG₁, PMDG₂ and MPMDG₂. The bands at 1459 cm^{-1} and 1724 cm^{-1} are assigned to OH in-plane deformation vibration and C=O stretching of COOH group, respectively [20], which enhance significantly in MPMDG₂ spectrum, indicating that more COOH groups are formed by introduction of OA-Fe₃O₄. The adsorption peak observed at 913 cm^{-1} in PMDG₁ spectrum, identified as the stretching vibration of epoxy group, weakens obviously in PMDG₂ and MPMDG₂ spectrums, while the stretching vibration of OH group (near 3500 cm^{-1}) becomes stronger in PMDG₂ and MPMDG₂ spectrums [21], indicating that epoxy groups are opened to release more OH group by acidic hydrolysis. Besides, the adsorption peak at 570 cm^{-1} , identified as the adsorption band of Fe–O, is observed in MPMDG₂ spectrum. These results indicate that carboxyl and hydroxyl groups were added to the resins, while OA-Fe₃O₄ was successfully embedded into the polymer matrix.

3.2. Adsorption of metal ions from aqueous solutions

3.2.1. Effect of solution pH on adsorption

The pH is an important factor affecting the adsorption behavior of adsorbents because of its effect on the surface charge of the adsorbents and the conversion of the metal ion species [22]. In this work, the adsorption capacities (Q) of HMIs onto MPMDG₂ microspheres were studied as a function of pH. The initial pH value was kept between 2.0 and 6.0. The results are depicted in Fig. 5, showing that the maximum adsorption for metal ions are 133.2 mg/g for Cr(III), 130.1 mg/g for Pb(II), 61.8 mg/g for Cu(II) and 21.1 mg/g for Cd(II), occurred at pH 6.0. As pH increases, weak acid ($-\text{COOH}$) can ionize to give more H^+ , and the adsorbent surface becomes more negatively charged, consequently facilitating the adsorption of metal ions and ion exchange. On the other hand, a high pH, e.g. higher than 6.0, could lead to the formation of metallic hydroxides e.g. $\text{Cu}(\text{OH})_2$ and $\text{Cr}(\text{OH})_3$ [20]. Therefore, to eliminate the effect of metal precipitation, all further adsorption experiments were performed at the pH value of 5.5.

3.2.2. Effect of contact time on adsorption

The adsorption of HMIs on MPMDG₂ as a function of contact time was studied at 25°C . The adsorption capacities increase with prolonging contact time shown in Fig. 6. The equilibrium adsorption

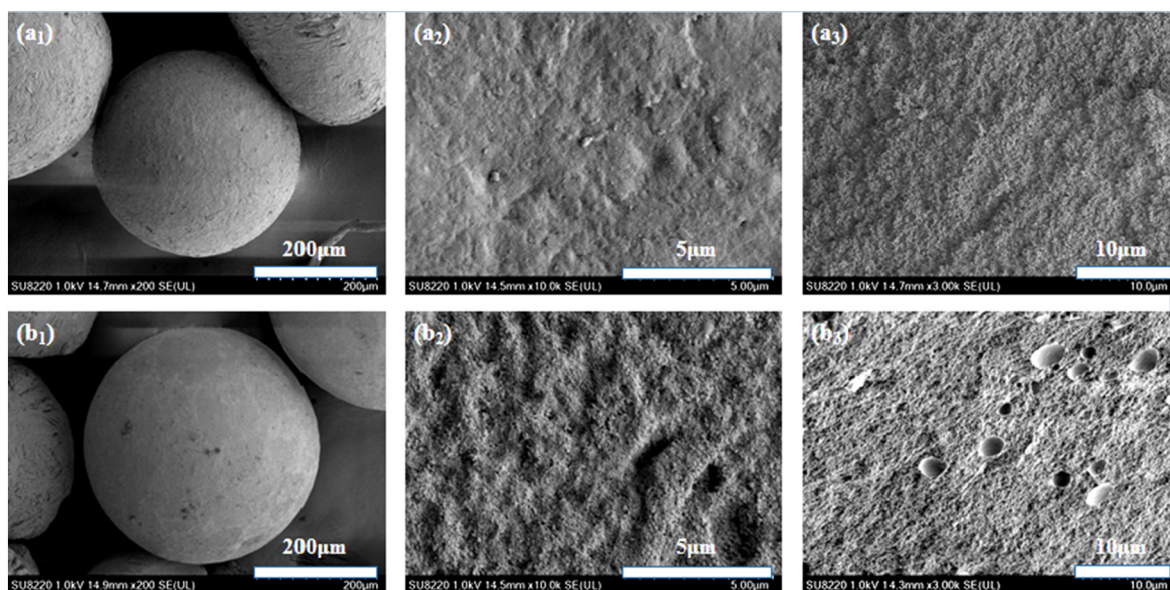


Fig. 3. SEM images of MPMDG₁ (a₁, a₂, a₃) and MPMDG₂ (b₁, b₂, b₃), a₂ and b₂: the surface of the beads; a₃ and b₃: the cross section of the beads.

capacities of four metal ions are 131.9 mg/g for Cr(III), 99.1 mg/g for Pd(II), 45.6 mg/g for Cu(II) and 13.5 mg/g for Cd(II). The resin exhibits obvious selective adsorption to Cr(III) and Pd(II) ions. It is found that 86% of Cd(II) and Cu(II), 76% of Cr(III) and 66% of Pb(II) (compared to the equilibrium adsorption value) were removed within first 2 min of contact time. This indicates that the initial rates of adsorption of Cd(II) and Cu(II) are faster than the adsorptions of Cr(III) and Pb(II). The rapid adsorption is related to the availability of active sites on adsorbent surface and the mesoporous property of the MPMDG₂. The mesoporous structure of resins could help to decrease the diffusion resistance and shorten the adsorption time of metal ions. The preferential adsorption of Cr(III) and Pb(II) in comparison to Cu(II) and Cd(II) is related to the valence state and the hydrated ionic radius of these cations [23]. Cr(III) has higher positive charge than other divalent ions, thus possessing greater ion exchange capacity. For three divalent ions, the order of hydrated ionic radius is Cd(II) > Cu(II) > Pb(II), consequently, the order of ion exchange capacity is Pb(II) > Cu(II) > Cd(II). Besides, different complexing capacities of metal ions with the COOH and OH groups on the surface of MPMDG₂ could affect the adsorption of metal ions. Jing's research revealed that Cu(II) and Pb(II) can coordinate with O atom in resin, whereas Cd(II) seldom coordinates with N or O atom in medium-hard lewis base because Cd(II) belongs to soft lewis acid (based on Brønsted-Lowry acid-base theory) [24]. This would be responsible for the less adsorption of Cd(II) ions than the other metal ions on the resin.

3.2.3. Effect of the initial concentration of metal ions

Fig. 7 shows the effect of metal ions initial concentrations on the adsorption capacity (Q) and removal efficiency (E) of HMIs. As the initial concentration increases, the Q values for four metal ions increase linearly, whereas the E values more or less decrease with the exception of Cr(III) ion. The maximum adsorption capacities are 152.6 mg/g for Cr(III), 108.9 mg/g for Pb(II), 94.5 mg/g for Cu(II) and 35.6 mg/g for Cd(II), obtained at an initial concentration of 230 mg/L. The E values

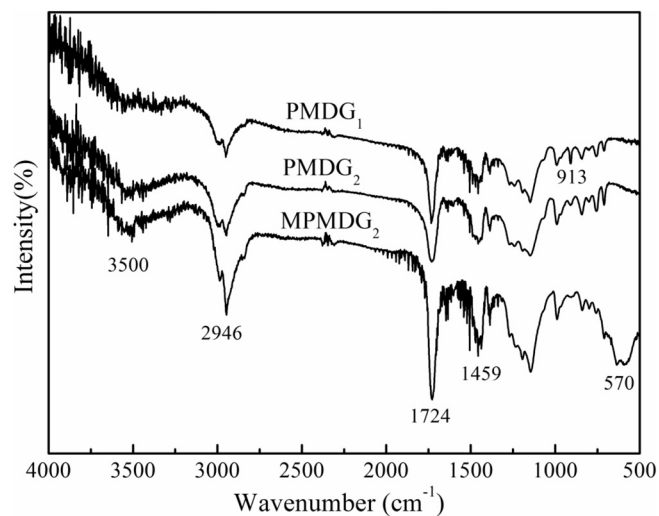


Fig. 4. FTIR spectra of PMDG₁, PMDG₂ and MPMDG₂.

for Cr(III) retain nearly 100%, while E values for Cu(II) and Pb(II) exceed 60% at the tested initial concentration range, indicating that the MPMDG₂ is an effective adsorbent for selective removal of Cr(III), Cu(II) and Pb(II) ions from wastewater. The adsorption selectivity for Cr(III), Cu(II) and Pb(II) would be attributed to the fact that the functional groups (COOH and OH) on the MPMDG₂ have a higher affinity to the three ions [22], and the three metal ions have a stronger ion-exchange capacity. The Q values for four metal ions at initial concentration of 200 mg/L are all much higher than those obtained on the lignin-based ion-exchange resins reported in the previous literature [20]. Therefore, MPMDG₂ would be a good candidate as an adsorbent for removing Cr(III), Pb(II) and Cu(II) from wastewater.

Table 1

Physicochemical properties of MPMDG₁ and MPMDG₂.

Resins	Specific surface area (m ² /g)	Average pore diameter (nm)	Total pore volume (cm ³ /g)	Functional groups
MPMDG ₁	307	4.0	0.31	–COOH
MPMDG ₂	281	4.6	0.33	–COOH, –OH

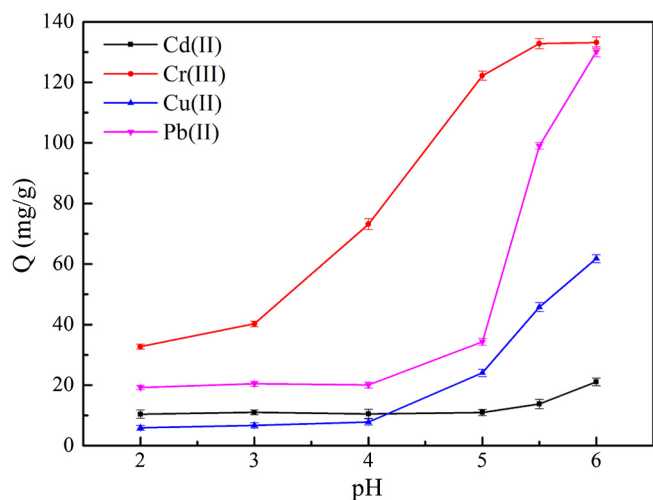


Fig. 5. Effect of pH on adsorption of HMIs on the MPMDG₂ (initial concentrations of Cu(II) and Cd(II): 100 mg/L; Cr(III) and Pb(II): 200 mg/L).

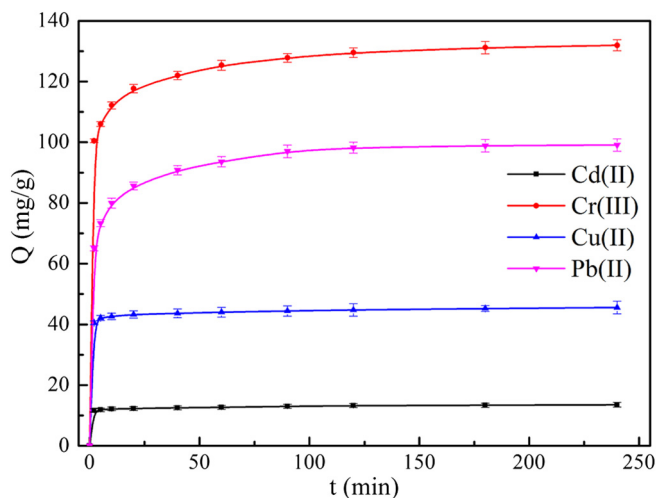


Fig. 6. Effect of contact time on adsorption of HMIs on MPMDG₂.

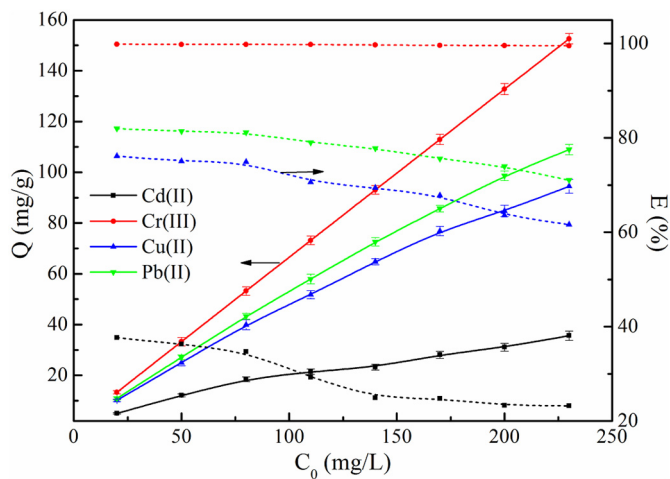


Fig. 7. Effect of the metal ions initial concentrations on the adsorption capacity and removal efficiency of HMIs.

3.3. Analysis of adsorption kinetics

In order to analyze the adsorption kinetics of HMIs, several kinetic models were used to test the experimental data. The pseudo-first-order kinetic equation is given as Eq. (4) [25]:

$$\log(Q_e - Q_t) = \log Q_e - \frac{k_1 t}{2.303} \quad (4)$$

and the pseudo-second-order equation is expressed as follows [26]:

$$\frac{t}{Q_t} = \frac{1}{k_2 Q_e^2} + \frac{t}{Q_e} \quad (5)$$

where Q_e and Q_t are the adsorption capacities at equilibrium and time t (min) (mg/g), respectively, k_1 is the pseudo-first-order rate constant (1/min), and k_2 is the pseudo-second-order rate constant (mg/(g min)).

Both pseudo-first-order and pseudo-second-order kinetic models cannot explain the diffusion mechanism in adsorption process. Therefore, the intraparticle diffusion model described as Eq. (6) was used to analyze the diffusion behavior [27]:

$$Q_t = k_p t^{0.5} \quad (6)$$

where k_p is the rate constant (mg/(g·min^{1/2})), which can be obtained from the plot of Q_t against $t^{0.5}$.

The data for the adsorption capacity with contact time (Fig. 6) were tested against the above three kinetic models. The results are presented in Fig. 8. The regression coefficient (R^2) values, the predicted Q_e values, and the rate constants of the three kinetic models (k_1 , k_2 , and k_p) are listed in Table 2. It can be clearly seen that the values of R^2 for the pseudo-second-order are higher than those of the other kinetic models. Moreover, the Q_e values for four metal ions predicted by the pseudo-second-order model are very close to the experimental values (Q_{exp}). Therefore, the adsorption of metal ions on MPMDG₂ can be best described by the pseudo-second-order kinetic model, implying that the adsorption of metal ions on the obtained resin is an ion exchange process [15].

Based on the curves of Q_t and $t^{0.5}$ (Fig. 8c), the adsorption behaviors of Cr(III) and Pb(II) can be described by three linear portions. The first portion with a steep slope, indicates a rapid diffusion stage, in which the metal ions are rapidly absorbed by the surface negatively charged groups ($-\text{COO}^-$ and $-\text{O}^-$). In the second portion, the metal ions gradually enter the inside of the resin matrix and are adsorbed onto the inner active sites. The third portion, where the slope is close to zero, indicates that the equilibrium state is reached. Since the straight line for the first portion does not pass through the origin, intraparticle diffusion is not the rate-determining step for ion adsorption [14]. The film diffusion of the metal ions (through the boundary layer that surrounds the resin microspheres) likely controls the adsorption process [28]. The diffusion rate constant k_p for Cr(III) and Pb(II) are higher than that for Cu(II) and Cd(II), which may be due to the higher affinity to MPMDG₂ and ion-exchange capacity of the two metal ions.

3.4. Analysis of adsorption isotherms

Adsorption isotherm analysis is important to describe the relationship between equilibrium adsorption capability and equilibrium concentration of the metal ions in solution. In order to better evaluate the adsorption capacity of an adsorbent and comprehend the interaction between the adsorbent and adsorbate, the Langmuir and Freundlich isotherm models were applied to simulate the experimental data.

The Langmuir isotherm is given as [29],

$$\frac{C_e}{Q_e} = \frac{C_e}{Q_m} + \frac{1}{Q_m b} \quad (7)$$

where C_e is the residual concentration of metal ions in solution (mg/L), Q_m is the maximum adsorption capacity of metal ions (mg/g), and b is the Langmuir constant related to the affinity of the binding sites (L/

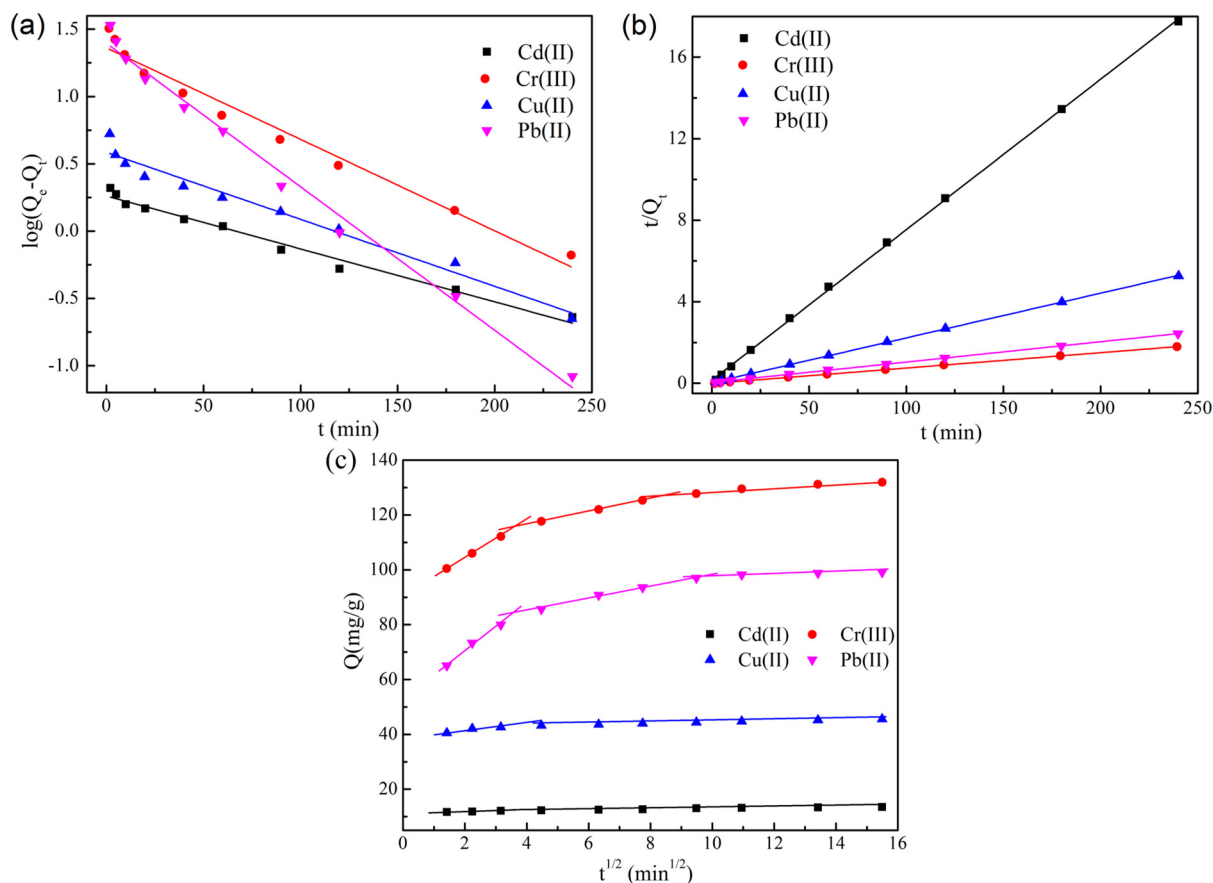


Fig. 8. Adsorption kinetics of Cu(II), Cd(II), Pb(II) and Cr(III) on MPMDG₂, (a) pseudo-first-order model, (b) pseudo-second-order model, (c) intraparticle diffusion plot.

mg).

The Freundlich model can be described by Eq. (7) [30],

$$Q_e = K_F C_e^{1/n} \quad (8)$$

or its logarithmic form:

$$\log Q_e = \log K_F + \frac{1}{n} \log C_e \quad (9)$$

where K_F is a constant relating to the adsorption capacity (mg/g) and $1/n$ is an empirical constant related to the adsorption intensity. The values of K_F and $1/n$ are calculated from the slope and intercept of the plot of $\log Q_e$ versus $\log C_e$. The $1/n$ values of the Freundlich model can indicate the type of isotherm. For $0 < 1/n < 1$, the adsorption of adsorbate onto adsorbent is favorable. A higher value of n means stronger adsorption intensity.

The calculated values of the Langmuir and Freundlich equation's parameters are presented in Table 3. It can be seen that the linear coefficients of determination (R^2) for the Langmuir isotherm model for Cr(III), Pb(II) and Cu(II) are slightly higher than that for Freundlich isotherm, indicating that Langmuir model is more suitable for

Table 3

Adsorption isotherm parameters for Cu(II), Cd(II), Pb(II) and Cr(III) on MPMDG₂.

Metal ions	Q_{exp}	Langmuir				Freundlich		
		b	Q_m	R^2		K_F	$1/n$	R^2
Cd(II)	13.75	0.03	5.97	0.95146		0.99	0.70	0.97075
Cr(III)	132.60	2.26	205.76	0.98526		160.18	0.65	0.97674
Cu(II)	45.80	0.01	173.01	0.98347		3.54	0.76	0.98021
Pb(II)	99.19	0.02	219.30	0.98566		4.49	0.79	0.98344

describing the adsorption of the three ions onto MPMDG₂. The adsorption is considered to be a monomolecular layer adsorption process [15]. The experimental adsorption capacities for Cr(III), Cu(II) and Pb(II) are much lower than the theoretical maximum adsorption capacities. It may be attributed to the disturbance from coexisting ions. Freundlich model is more suitable for describing the adsorption behavior of Cd(II), suggesting that the adsorption of Cd(II) on the resin would be a multimolecular layer adsorption process. The competitive

Table 2

Kinetic adsorption parameters for Cu(II), Cd(II), Pb(II) and Cr(III) on MPMDG₂.

Metal ions	Pseudo-first-order model				Pseudo-second-order model			Intraparticle diffusion	
	Q_{exp}	$k_1(10^{-2})$	Q_e	R^2	$k_2(10^{-2})$	Q_e	R^2	k_p	R^2
Cd(II)	13.8	0.91	1.8	0.98167	3.58	13.5	0.99697	0.44	0.25808
Cr(III)	132.6	1.57	23.2	0.97315	0.33	132.6	0.99792	4.85	0.36486
Cu(II)	45.8	1.14	3.8	0.96880	1.83	45.6	0.99897	1.38	0.20348
Pb(II)	99.2	2.45	25.0	0.98869	0.37	100.1	0.99894	4.09	0.46713

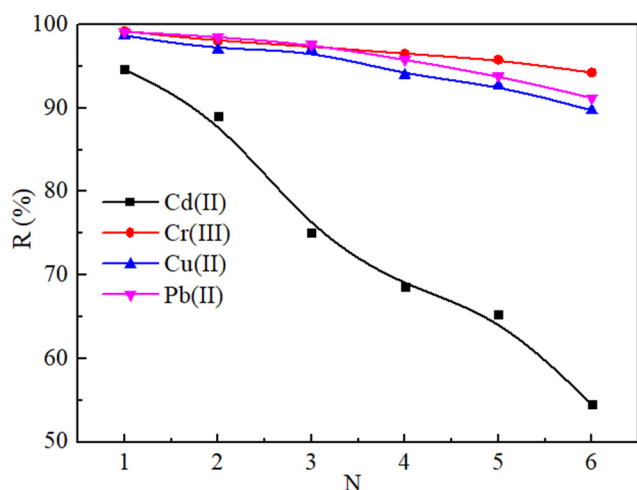


Fig. 9. Desorption-adsorption cycle efficiency (R) of MPMDG₂ for different metal ions.

adsorption of Cr(III), Cu(II) and Pb(II) onto the resin could change the surface adsorption energy of resin, consequently retarding the adsorption of Cd(II) ion. In addition, the $1/n$ values for the Freundlich adsorption isotherms of four metal ions are lower than 1, implying that the adsorption of the four metal ions on MPMDG₂ is favorable.

Based on the above analysis, the authors assume that the adsorption of metal ions by resin proceeds via several stages: the transfer of metal ions from the bulk solution to the liquid-solid interface (occurs very quickly); the diffusion of metal ions through the boundary layer that surrounds the resin particles (the rate-determining step); the diffusion of metal ions from the surface to the interior of the resin particles (occurs very quickly); and the adsorption of metal ions to the active sites on the resin surface (occurs very quickly) [31]. The driving forces for adsorption include physical and chemical factors such as diffusion caused by the concentration difference of metal ions between the solution and the interior of the resin, van der Waals forces between metal ions and resin microspheres, electrostatic force between the metal ions and the electronegative group ($-\text{COO}^-$) of the resin, ion exchange, and chemistry chelating between metal ions and functional groups ($-\text{COOH}$, $-\text{OH}$). Therefore, the adsorption of HMIs on the MPMDG₂ could be a combined effect of both chemisorption and physisorption, with ion exchange and complexation as the main processes.

3.5. Regeneration of cation-exchange resins

The use of an adsorbent in the adsorption process depends not only on the adsorptive capacity but also on how well the adsorbent can be regenerated and used again. Fig. 9 shows the adsorption-desorption cycle efficiency of MPMDG₂ for the different metal ions after each adsorption/desorption cycle. As can be seen, even after six cycles, the MPMDG₂ still exhibits high adsorption efficiencies (higher than 90%) for Cu(II), Pb(II) and Cr(III), indicating that the MPMDG₂ resin has good reusability. The MPMDG₂ can be easily recovered from wastewater by applying an external magnetic field, suggesting its potential for application in the treatment of actual HMI wastewater.

4. Conclusions

In this study, magnetic resin microspheres called MPMDG₂ were prepared by suspension polymerization combined with basic and acidic hydrolysis. The adsorption property of resin to Cr(III), Cu(II), Cd(II) and Pb(II) ions was investigated. Structural and morphological analyses revealed that the resulting resin had higher specific surface area and mesoporous structure. The resin contained about 9.2 wt% Fe_3O_4 , exhibiting a high magnetic response. The resin showed selective

adsorption to Cr(III), Pb(II) and Cu(II). The adsorption processes attained equilibrium within 120 min for Cr(III) and Pd(II) and 20 min for Cd(II) and Cu(II), and were well described by a pseudo-second-order kinetic model. The equilibrium adsorption data fitted the Langmuir isotherm model better than the Freundlich model. The adsorption mechanisms of metal ions mainly involved ion exchange and chemistry chelating. The superior magnetic response, regeneration as well as adsorption selectivity of the MPMDG₂ make it a good candidate as an adsorbent for removing Cr(III), Pb(II) and Cu(II) from wastewater.

Declaration of competing interest

The authors declare no competing financial interest.

Acknowledgments

This work was supported by the National Natural Science Foundation of China (21566004), and the Scientific Research Foundation of Guangxi University (XGZ120081, XTZ140787, XJPZ160713).

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